

SAFETY DATA SHEET

Lactose Monohydrate (Pharmaceutical Excipient Grade)

SECTION 1: Identification

Product Name: Lactose Monohydrate, USP/IP/Ph.Eur. Grade

Use: Pharmaceutical excipient (tablet diluent/filler)

Supplier (fictional, for demo): Meridian Excipients Supply Co., Ahmedabad, India

Emergency Contact (fictional): +91-79-5555-0123 (24 hr)

SECTION 2: Hazard(s) Identification

GHS Classification: Not classified as hazardous under GHS. Not flammable, not a skin or eye irritant under normal handling. Fine powder may be a dust nuisance; avoid prolonged inhalation of dust.

SECTION 3: Composition / Information on Ingredients

Lactose Monohydrate (CAS 64044-51-5), $\geq 99.0\%$ w/w. No other hazardous components present above reportable thresholds.

SECTION 4: First-Aid Measures

Inhalation: Remove to fresh air if dust discomfort occurs. Skin Contact: Wash with water if irritation occurs. Eye Contact: Flush with water for several minutes. Ingestion: Not expected to be harmful; seek medical advice if large quantity ingested or if lactose intolerant.

SECTION 5: Fire-Fighting Measures

Non-flammable under normal conditions. Fine powder may present a dust explosion risk in high concentration with an ignition source; use water spray, CO₂, or dry chemical as appropriate.

SECTION 6: Accidental Release Measures

Sweep or vacuum spilled material to avoid dust generation. Avoid dry sweeping. Place in appropriate waste container. Wash spill area with water.

SECTION 7: Handling and Storage

Handle in a well-ventilated area with dust control (local exhaust) as needed. Store in tightly closed containers in a cool, dry place away from moisture, per site SOP-WH-004 (Storage of Excipients).

SECTION 8: Exposure Controls / Personal Protection

Engineering Controls: Local exhaust ventilation/dust extraction at dispensing points. PPE: Dust mask, safety glasses, and gloves during dispensing per SOP-PRD-003.

SECTION 9: Physical and Chemical Properties

Appearance: White crystalline powder. Odor: Odorless. Solubility: Soluble in water. Melting Point: $\sim 201\text{-}202\text{ }^{\circ}\text{C}$ (decomposes). pH (10% aqueous solution): 4.5-7.0.

SECTION 10: Stability and Reactivity

Stable under normal storage conditions. Hygroscopic under high humidity. Avoid strong oxidizing agents. No hazardous decomposition products expected under normal use.

SECTION 11: Toxicological Information

Low acute toxicity. May cause gastrointestinal discomfort in individuals with lactose intolerance if ingested in large amounts. No known carcinogenic, mutagenic, or reproductive toxicity classification.

SECTION 12: Ecological Information

Readily biodegradable. Not expected to bioaccumulate. No significant aquatic toxicity data indicating hazard at typical release volumes.

SECTION 13: Disposal Considerations

Dispose of in accordance with local regulations. Non-hazardous waste; may be disposed of as general solid waste unless contaminated with a hazardous substance during processing.

SECTION 14: Transport Information

Not classified as a dangerous good under ADR/IATA/IMDG. No special transport precautions required.

SECTION 15: Regulatory Information

Complies with USP/NF, Ph.Eur., and IP monograph requirements for pharmaceutical-grade lactose monohydrate. Not subject to SARA Title III or similar hazardous-substance reporting thresholds.

SECTION 16: Other Information

This SDS is a synthetic example generated for AI hackathon prototype testing and does not represent an official manufacturer safety data sheet. Revision Date: 20-Jan-2026.

Confidential — Synthetic training data generated for AI hackathon prototype. Not an official safety data sheet.